

1 Instances Precedence (CPO)

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
1 20 3 10 rep 1.txt	20	3	10	Solution	30.09	741	631.00	14.84
2 20 3 10 rep 2.txt	20	3	10	Solution	30.01	715	674.00	5.73
3 20 3 10 rep 3.txt	20	3	10	Optimal	22.15	488	488.00	0.00
4 20 3 10 rep 4.txt	20	3	10	Solution	30.01	742	669.00	9.84
5 20 3 10 rep 5.txt	20	3	10	Solution	30.01	880	867.00	1.48
6 20 3 10 rep 6.txt	20	3	10	Solution	30.01	670	618.00	7.76
7 20 3 10 rep 7.txt	20	3	10	Solution	30.01	718	668.00	6.96
8 20 3 10 rep 8.txt	20	3	10	Solution	30.01	813	754.00	7.26
9 20 3 10 rep 9.txt	20	3	10	Optimal	28.58	576	576.00	0.00
10 20 3 10 rep 10.txt	20	3	10	Solution	30.01	784	703.00	10.33
11 20 3 20 rep 1.txt	20	3	20	Optimal	7.67	400	400.00	0.00
12 20 3 20 rep 2.txt	20	3	20	Optimal	1.01	355	355.00	0.00
13 20 3 20 rep 3.txt	20	3	20	Solution	30.01	421	414.00	1.66
14 20 3 20 rep 4.txt	20	3	20	Solution	30.01	376	341.00	9.31
15 20 3 20 rep 5.txt	20	3	20	Optimal	4.34	344	344.00	0.00
16 20 3 20 rep 6.txt	20	3	20	Optimal	1.94	377	377.00	0.00
17 20 3 20 rep 7.txt	20	3	20	Solution	30.01	368	345.00	6.25
18 20 3 20 rep 8.txt	20	3	20	Optimal	14.50	412	412.00	0.00
19 20 3 20 rep 9.txt	20	3	20	Optimal	15.58	294	294.00	0.00
20 20 3 20 rep 10.txt	20	3	20	Optimal	4.29	349	349.00	0.00
21 20 3 30 rep 1.txt	20	3	30	Optimal	2.13	335	335.00	0.00
22 20 3 30 rep 2.txt	20	3	30	Optimal	7.03	295	295.00	0.00
23 20 3 30 rep 3.txt	20	3	30	Optimal	2.73	301	301.00	0.00
24 20 3 30 rep 4.txt	20	3	30	Optimal	0.75	289	289.00	0.00
25 20 3 30 rep 5.txt	20	3	30	Optimal	3.84	292	292.00	0.00
26 20 3 30 rep 6.txt	20	3	30	Optimal	5.38	287	287.00	0.00
27 20 3 30 rep 7.txt	20	3	30	Optimal	6.68	345	345.00	0.00
28 20 3 30 rep 8.txt	20	3	30	Optimal	2.85	322	322.00	0.00
29 20 3 30 rep 9.txt	20	3	30	Optimal	4.41	300	300.00	0.00
30 20 3 30 rep 10.txt	20	3	30	Optimal	0.19	245	245.00	0.00
31 20 3 50 rep 1.txt	20	3	50	Optimal	0.02	253	253.00	0.00
32 20 3 50 rep 2.txt	20	3	50	Optimal	0.10	237	237.00	0.00
33 20 3 50 rep 3.txt	20	3	50	Optimal	0.02	241	241.00	0.00
34 20 3 50 rep 4.txt	20	3	50	Optimal	0.11	231	231.00	0.00
35 20 3 50 rep 5.txt	20	3	50	Optimal	0.02	272	272.00	0.00
36 20 3 50 rep 6.txt	20	3	50	Optimal	0.02	216	216.00	0.00
37 20 3 50 rep 7.txt	20	3	50	Optimal	0.02	224	224.00	0.00
38 20 3 50 rep 8.txt	20	3	50	Optimal	0.01	264	264.00	0.00
39 20 3 50 rep 9.txt	20	3	50	Optimal	0.05	234	234.00	0.00
40 20 3 50 rep 10.txt	20	3	50	Optimal	0.02	231	231.00	0.00
41 20 5 10 rep 1.txt	20	5	10	Solution	30.01	897	829.00	7.58
42 20 5 10 rep 2.txt	20	5	10	Solution	30.01	869	759.00	12.66

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
43 20 5 10 rep 3.txt	20	5	10	Solution	30.01	885	798.00	9.83
44 20 5 10 rep 4.txt	20	5	10	Solution	30.01	849	729.00	14.13
45 20 5 10 rep 5.txt	20	5	10	Solution	30.01	866	767.00	11.43
46 20 5 10 rep 6.txt	20	5	10	Solution	30.01	822	683.00	16.91
47 20 5 10 rep 7.txt	20	5	10	Solution	30.01	886	759.00	14.33
48 20 5 10 rep 8.txt	20	5	10	Solution	30.01	845	759.00	10.18
49 20 5 10 rep 9.txt	20	5	10	Solution	30.01	844	784.00	7.11
50 20 5 10 rep 10.txt	20	5	10	Solution	30.01	1072	940.00	12.31
51 20 5 20 rep 1.txt	20	5	20	Solution	30.01	499	455.00	8.82
52 20 5 20 rep 2.txt	20	5	20	Optimal	3.97	434	434.00	0.00
53 20 5 20 rep 3.txt	20	5	20	Solution	30.01	516	460.00	10.85
54 20 5 20 rep 4.txt	20	5	20	Solution	30.01	387	372.00	3.88
55 20 5 20 rep 5.txt	20	5	20	Optimal	6.16	470	470.00	0.00
56 20 5 20 rep 6.txt	20	5	20	Solution	30.01	502	460.00	8.37
57 20 5 20 rep 7.txt	20	5	20	Optimal	6.04	459	459.00	0.00
58 20 5 20 rep 8.txt	20	5	20	Optimal	6.79	496	496.00	0.00
59 20 5 20 rep 9.txt	20	5	20	Optimal	4.03	518	518.00	0.00
60 20 5 20 rep 10.txt	20	5	20	Solution	30.01	480	427.00	11.04
61 20 5 30 rep 1.txt	20	5	30	Optimal	4.03	386	386.00	0.00
62 20 5 30 rep 2.txt	20	5	30	Optimal	5.17	451	451.00	0.00
63 20 5 30 rep 3.txt	20	5	30	Optimal	0.22	367	367.00	0.00
64 20 5 30 rep 4.txt	20	5	30	Solution	30.01	425	419.00	1.41
65 20 5 30 rep 5.txt	20	5	30	Optimal	1.82	380	380.00	0.00
66 20 5 30 rep 6.txt	20	5	30	Optimal	1.47	401	401.00	0.00
67 20 5 30 rep 7.txt	20	5	30	Optimal	0.02	431	431.00	0.00
68 20 5 30 rep 8.txt	20	5	30	Optimal	0.72	404	404.00	0.00
69 20 5 30 rep 9.txt	20	5	30	Optimal	5.11	435	435.00	0.00
70 20 5 30 rep 10.txt	20	5	30	Optimal	13.37	436	436.00	0.00
71 20 5 50 rep 1.txt	20	5	50	Optimal	0.02	312	312.00	0.00
72 20 5 50 rep 2.txt	20	5	50	Optimal	0.06	323	323.00	0.00
73 20 5 50 rep 3.txt	20	5	50	Optimal	0.01	353	353.00	0.00
74 20 5 50 rep 4.txt	20	5	50	Optimal	0.02	348	348.00	0.00
75 20 5 50 rep 5.txt	20	5	50	Optimal	0.02	371	371.00	0.00
76 20 5 50 rep 6.txt	20	5	50	Optimal	0.02	425	425.00	0.00
77 20 5 50 rep 7.txt	20	5	50	Optimal	0.02	349	349.00	0.00
78 20 5 50 rep 8.txt	20	5	50	Optimal	0.02	398	398.00	0.00
79 20 5 50 rep 9.txt	20	5	50	Optimal	0.02	398	398.00	0.00
80 20 5 50 rep 10.txt	20	5	50	Optimal	0.01	386	386.00	0.00
81 20 10 10 rep 1.txt	20	10	10	Solution	30.01	1255	964.00	23.19
82 20 10 10 rep 2.txt	20	10	10	Solution	30.01	1220	997.00	18.28
83 20 10 10 rep 3.txt	20	10	10	Solution	30.01	1224	1023.00	16.42
84 20 10 10 rep 4.txt	20	10	10	Solution	30.01	1210	1112.00	8.10
85 20 10 10 rep 5.txt	20	10	10	Solution	30.01	1283	1045.00	18.55
86 20 10 10 rep 6.txt	20	10	10	Solution	30.01	1256	1063.00	15.37
87 20 10 10 rep 7.txt	20	10	10	Solution	30.01	1225	1014.00	17.22

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
88 20 10 10 rep 8.txt	20	10	10	Solution	30.01	1269	1086.00	14.42
89 20 10 10 rep 9.txt	20	10	10	Solution	30.01	1193	1000.00	16.18
90 20 10 10 rep 10.txt	20	10	10	Solution	30.01	1214	978.00	19.44
91 20 10 20 rep 1.txt	20	10	20	Solution	30.01	741	695.00	6.21
92 20 10 20 rep 2.txt	20	10	20	Solution	30.01	857	722.00	15.75
93 20 10 20 rep 3.txt	20	10	20	Optimal	5.24	699	699.00	0.00
94 20 10 20 rep 4.txt	20	10	20	Solution	30.01	851	736.00	13.51
95 20 10 20 rep 5.txt	20	10	20	Solution	30.01	722	699.00	3.19
96 20 10 20 rep 6.txt	20	10	20	Solution	30.01	873	788.00	9.74
97 20 10 20 rep 7.txt	20	10	20	Solution	30.01	791	723.00	8.60
98 20 10 20 rep 8.txt	20	10	20	Solution	30.01	813	730.00	10.21
99 20 10 20 rep 9.txt	20	10	20	Solution	30.01	814	704.00	13.51
100 20 10 20 rep 10.txt	20	10	20	Solution	30.01	745	732.00	1.74
101 20 10 30 rep 1.txt	20	10	30	Optimal	0.08	730	730.00	0.00
102 20 10 30 rep 2.txt	20	10	30	Optimal	0.04	780	780.00	0.00
103 20 10 30 rep 3.txt	20	10	30	Optimal	0.68	614	614.00	0.00
104 20 10 30 rep 4.txt	20	10	30	Optimal	0.10	726	726.00	0.00
105 20 10 30 rep 5.txt	20	10	30	Solution	30.01	668	658.00	1.50
106 20 10 30 rep 6.txt	20	10	30	Optimal	0.93	639	639.00	0.00
107 20 10 30 rep 7.txt	20	10	30	Optimal	0.09	728	728.00	0.00
108 20 10 30 rep 8.txt	20	10	30	Optimal	0.04	763	763.00	0.00
109 20 10 30 rep 9.txt	20	10	30	Optimal	0.52	645	645.00	0.00
110 20 10 30 rep 10.txt	20	10	30	Optimal	0.09	692	692.00	0.00
111 20 10 50 rep 1.txt	20	10	50	Optimal	0.02	668	668.00	0.00
112 20 10 50 rep 2.txt	20	10	50	Optimal	0.02	740	740.00	0.00
113 20 10 50 rep 3.txt	20	10	50	Optimal	0.02	701	701.00	0.00
114 20 10 50 rep 4.txt	20	10	50	Optimal	0.02	693	693.00	0.00
115 20 10 50 rep 5.txt	20	10	50	Optimal	0.02	651	651.00	0.00
116 20 10 50 rep 6.txt	20	10	50	Optimal	0.02	641	641.00	0.00
117 20 10 50 rep 7.txt	20	10	50	Optimal	0.02	654	654.00	0.00
118 20 10 50 rep 8.txt	20	10	50	Optimal	0.02	673	673.00	0.00
119 20 10 50 rep 9.txt	20	10	50	Optimal	0.02	708	708.00	0.00
120 20 10 50 rep 10.txt	20	10	50	Optimal	0.02	731	731.00	0.00
121 30 3 10 rep 1.txt	30	3	10	Solution	30.01	937	887.00	5.34
122 30 3 10 rep 2.txt	30	3	10	Solution	30.01	1242	1114.00	10.31
123 30 3 10 rep 3.txt	30	3	10	Solution	30.01	917	864.00	5.78
124 30 3 10 rep 4.txt	30	3	10	Solution	30.01	1202	1031.00	14.23
125 30 3 10 rep 5.txt	30	3	10	Solution	30.01	1105	996.00	9.86
126 30 3 10 rep 6.txt	30	3	10	Solution	30.01	1198	1069.00	10.77
127 30 3 10 rep 7.txt	30	3	10	Solution	30.01	1221	1121.00	8.19
128 30 3 10 rep 8.txt	30	3	10	Solution	30.01	1181	998.00	15.50
129 30 3 10 rep 9.txt	30	3	10	Solution	30.01	993	972.00	2.11
130 30 3 10 rep 10.txt	30	3	10	Solution	30.01	1039	955.00	8.08
131 30 3 20 rep 1.txt	30	3	20	Solution	30.01	550	526.00	4.36
132 30 3 20 rep 2.txt	30	3	20	Solution	30.01	571	539.00	5.60

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
133 30 3 20 rep 3.txt	30	3	20	Solution	30.01	468	438.00	6.41
134 30 3 20 rep 4.txt	30	3	20	Solution	30.01	614	594.00	3.26
135 30 3 20 rep 5.txt	30	3	20	Solution	30.01	451	419.00	7.10
136 30 3 20 rep 6.txt	30	3	20	Solution	30.01	499	487.00	2.40
137 30 3 20 rep 7.txt	30	3	20	Solution	30.01	493	474.00	3.85
138 30 3 20 rep 8.txt	30	3	20	Solution	30.01	550	535.00	2.73
139 30 3 20 rep 9.txt	30	3	20	Solution	30.01	498	485.00	2.61
140 30 3 20 rep 10.txt	30	3	20	Solution	30.01	468	451.00	3.63
141 30 3 30 rep 1.txt	30	3	30	Solution	30.01	437	425.00	2.75
142 30 3 30 rep 2.txt	30	3	30	Solution	30.01	325	316.00	2.77
143 30 3 30 rep 3.txt	30	3	30	Solution	30.01	449	440.00	2.00
144 30 3 30 rep 4.txt	30	3	30	Solution	30.01	340	333.00	2.06
145 30 3 30 rep 5.txt	30	3	30	Solution	30.01	433	423.00	2.31
146 30 3 30 rep 6.txt	30	3	30	Solution	30.00	323	304.00	5.88
147 30 3 30 rep 7.txt	30	3	30	Solution	30.01	418	407.00	2.63
148 30 3 30 rep 8.txt	30	3	30	Solution	30.01	396	382.00	3.54
149 30 3 30 rep 9.txt	30	3	30	Solution	30.01	418	412.00	1.44
150 30 3 30 rep 10.txt	30	3	30	Solution	30.01	366	359.00	1.91
151 30 3 50 rep 1.txt	30	3	50	Optimal	0.09	252	252.00	0.00
152 30 3 50 rep 2.txt	30	3	50	Optimal	0.10	258	258.00	0.00
153 30 3 50 rep 3.txt	30	3	50	Optimal	0.04	294	294.00	0.00
154 30 3 50 rep 4.txt	30	3	50	Optimal	0.29	235	235.00	0.00
155 30 3 50 rep 5.txt	30	3	50	Optimal	21.81	253	253.00	0.00
156 30 3 50 rep 6.txt	30	3	50	Optimal	0.66	268	268.00	0.00
157 30 3 50 rep 7.txt	30	3	50	Optimal	2.60	252	252.00	0.00
158 30 3 50 rep 8.txt	30	3	50	Solution	30.01	260	257.00	1.15
159 30 3 50 rep 9.txt	30	3	50	Optimal	0.04	275	275.00	0.00
160 30 3 50 rep 10.txt	30	3	50	Solution	30.01	264	259.00	1.89
161 30 5 10 rep 1.txt	30	5	10	Solution	30.01	1244	1053.00	15.35
162 30 5 10 rep 2.txt	30	5	10	Solution	30.01	1201	1093.00	8.99
163 30 5 10 rep 3.txt	30	5	10	Solution	30.01	1201	1091.00	9.16
164 30 5 10 rep 4.txt	30	5	10	Solution	30.01	1134	1002.00	11.64
165 30 5 10 rep 5.txt	30	5	10	Solution	30.01	1330	1170.00	12.03
166 30 5 10 rep 6.txt	30	5	10	Solution	30.01	1121	907.00	19.09
167 30 5 10 rep 7.txt	30	5	10	Solution	30.01	1182	1099.00	7.02
168 30 5 10 rep 8.txt	30	5	10	Solution	30.01	1541	1370.00	11.10
169 30 5 10 rep 9.txt	30	5	10	Solution	30.01	1022	872.00	14.68
170 30 5 10 rep 10.txt	30	5	10	Solution	30.01	1217	1178.00	3.20
171 30 5 20 rep 1.txt	30	5	20	Solution	30.01	676	582.00	13.91
172 30 5 20 rep 2.txt	30	5	20	Solution	30.01	605	585.00	3.31
173 30 5 20 rep 3.txt	30	5	20	Solution	30.01	651	593.00	8.91
174 30 5 20 rep 4.txt	30	5	20	Solution	30.01	690	619.00	10.29
175 30 5 20 rep 5.txt	30	5	20	Solution	30.01	564	533.00	5.50
176 30 5 20 rep 6.txt	30	5	20	Solution	30.01	670	646.00	3.58
177 30 5 20 rep 7.txt	30	5	20	Solution	30.01	661	600.00	9.23

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
178 30 5 20 rep 8.txt	30	5	20	Solution	30.01	614	544.00	11.40
179 30 5 20 rep 9.txt	30	5	20	Solution	30.01	598	535.00	10.54
180 30 5 20 rep 10.txt	30	5	20	Solution	30.01	678	655.00	3.39
181 30 5 30 rep 1.txt	30	5	30	Solution	30.01	455	431.00	5.27
182 30 5 30 rep 2.txt	30	5	30	Solution	30.00	478	446.00	6.69
183 30 5 30 rep 3.txt	30	5	30	Solution	30.01	476	454.00	4.62
184 30 5 30 rep 4.txt	30	5	30	Solution	30.01	480	451.00	6.04
185 30 5 30 rep 5.txt	30	5	30	Solution	30.01	507	463.00	8.68
186 30 5 30 rep 6.txt	30	5	30	Solution	30.01	410	399.00	2.68
187 30 5 30 rep 7.txt	30	5	30	Solution	30.01	477	447.00	6.29
188 30 5 30 rep 8.txt	30	5	30	Solution	30.01	467	461.00	1.28
189 30 5 30 rep 9.txt	30	5	30	Solution	30.01	471	452.00	4.03
190 30 5 30 rep 10.txt	30	5	30	Solution	30.01	506	472.00	6.72
191 30 5 50 rep 1.txt	30	5	50	Optimal	28.10	366	366.00	0.00
192 30 5 50 rep 2.txt	30	5	50	Optimal	0.09	378	378.00	0.00
193 30 5 50 rep 3.txt	30	5	50	Optimal	5.33	378	378.00	0.00
194 30 5 50 rep 4.txt	30	5	50	Optimal	0.10	382	382.00	0.00
195 30 5 50 rep 5.txt	30	5	50	Optimal	7.25	365	365.00	0.00
196 30 5 50 rep 6.txt	30	5	50	Optimal	0.03	379	379.00	0.00
197 30 5 50 rep 7.txt	30	5	50	Optimal	4.90	394	394.00	0.00
198 30 5 50 rep 8.txt	30	5	50	Optimal	0.02	445	445.00	0.00
199 30 5 50 rep 9.txt	30	5	50	Optimal	0.46	360	360.00	0.00
200 30 5 50 rep 10.txt	30	5	50	Optimal	7.49	373	373.00	0.00
201 30 10 10 rep 1.txt	30	10	10	Solution	30.01	1697	1478.00	12.91
202 30 10 10 rep 2.txt	30	10	10	Solution	30.01	1479	1147.00	22.45
203 30 10 10 rep 3.txt	30	10	10	Solution	30.01	1577	1237.00	21.56
204 30 10 10 rep 4.txt	30	10	10	Solution	30.02	1630	1412.00	13.37
205 30 10 10 rep 5.txt	30	10	10	Solution	30.02	1621	1280.00	21.04
206 30 10 10 rep 6.txt	30	10	10	Solution	30.01	1626	1417.00	12.85
207 30 10 10 rep 7.txt	30	10	10	Solution	30.01	1583	1323.00	16.42
208 30 10 10 rep 8.txt	30	10	10	Solution	30.01	1542	1231.00	20.17
209 30 10 10 rep 9.txt	30	10	10	Solution	30.01	1660	1268.00	23.61
210 30 10 10 rep 10.txt	30	10	10	Solution	30.01	1608	1365.00	15.11
211 30 10 20 rep 1.txt	30	10	20	Solution	30.01	941	824.00	12.43
212 30 10 20 rep 2.txt	30	10	20	Solution	30.01	873	727.00	16.72
213 30 10 20 rep 3.txt	30	10	20	Solution	30.01	974	848.00	12.94
214 30 10 20 rep 4.txt	30	10	20	Solution	30.01	884	791.00	10.52
215 30 10 20 rep 5.txt	30	10	20	Solution	30.01	941	848.00	9.88
216 30 10 20 rep 6.txt	30	10	20	Solution	30.02	1000	890.00	11.00
217 30 10 20 rep 7.txt	30	10	20	Solution	30.01	970	861.00	11.24
218 30 10 20 rep 8.txt	30	10	20	Solution	30.01	928	862.00	7.11
219 30 10 20 rep 9.txt	30	10	20	Solution	30.02	897	783.00	12.71
220 30 10 20 rep 10.txt	30	10	20	Solution	30.01	945	803.00	15.03
221 30 10 30 rep 1.txt	30	10	30	Solution	30.02	753	714.00	5.18
222 30 10 30 rep 2.txt	30	10	30	Solution	30.01	771	742.00	3.76

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
223 30 10 30 rep 3.txt	30	10	30	Optimal	6.45	698	698.00	0.00
224 30 10 30 rep 4.txt	30	10	30	Solution	30.01	721	710.00	1.53
225 30 10 30 rep 5.txt	30	10	30	Solution	30.01	774	692.00	10.59
226 30 10 30 rep 6.txt	30	10	30	Solution	30.02	783	734.00	6.26
227 30 10 30 rep 7.txt	30	10	30	Solution	30.01	735	690.00	6.12
228 30 10 30 rep 8.txt	30	10	30	Optimal	0.88	791	791.00	0.00
229 30 10 30 rep 9.txt	30	10	30	Solution	30.01	751	707.00	5.86
230 30 10 30 rep 10.txt	30	10	30	Solution	30.01	783	754.00	3.70
231 30 10 50 rep 1.txt	30	10	50	Optimal	0.03	738	738.00	0.00
232 30 10 50 rep 2.txt	30	10	50	Optimal	0.02	646	646.00	0.00
233 30 10 50 rep 3.txt	30	10	50	Optimal	0.02	753	753.00	0.00
234 30 10 50 rep 4.txt	30	10	50	Optimal	0.08	652	652.00	0.00
235 30 10 50 rep 5.txt	30	10	50	Optimal	0.08	667	667.00	0.00
236 30 10 50 rep 6.txt	30	10	50	Optimal	0.02	687	687.00	0.00
237 30 10 50 rep 7.txt	30	10	50	Optimal	0.02	827	827.00	0.00
238 30 10 50 rep 8.txt	30	10	50	Optimal	0.02	692	692.00	0.00
239 30 10 50 rep 9.txt	30	10	50	Optimal	0.17	671	671.00	0.00
240 30 10 50 rep 10.txt	30	10	50	Optimal	0.02	764	764.00	0.00
241 50 3 10 rep 1.txt	50	3	10	Solution	30.01	1755	1623.00	7.52
242 50 3 10 rep 2.txt	50	3	10	Solution	30.01	1669	1473.00	11.74
243 50 3 10 rep 3.txt	50	3	10	Solution	30.01	1704	1586.00	6.92
244 50 3 10 rep 4.txt	50	3	10	Solution	30.11	1780	1686.00	5.28
245 50 3 10 rep 5.txt	50	3	10	Solution	30.02	1583	1504.00	4.99
246 50 3 10 rep 6.txt	50	3	10	Solution	30.02	1622	1539.00	5.12
247 50 3 10 rep 7.txt	50	3	10	Solution	30.02	1635	1548.00	5.32
248 50 3 10 rep 8.txt	50	3	10	Solution	30.01	1520	1436.00	5.53
249 50 3 10 rep 9.txt	50	3	10	Solution	30.02	1731	1667.00	3.70
250 50 3 10 rep 10.txt	50	3	10	Solution	30.01	1387	1339.00	3.46
251 50 3 20 rep 1.txt	50	3	20	Solution	30.02	748	714.00	4.55
252 50 3 20 rep 2.txt	50	3	20	Solution	30.02	869	850.00	2.19
253 50 3 20 rep 3.txt	50	3	20	Solution	30.01	783	760.00	2.94
254 50 3 20 rep 4.txt	50	3	20	Solution	30.01	727	712.00	2.06
255 50 3 20 rep 5.txt	50	3	20	Solution	30.02	757	735.00	2.91
256 50 3 20 rep 6.txt	50	3	20	Solution	30.01	836	797.00	4.67
257 50 3 20 rep 7.txt	50	3	20	Solution	30.01	828	813.00	1.81
258 50 3 20 rep 8.txt	50	3	20	Solution	30.01	797	779.00	2.26
259 50 3 20 rep 9.txt	50	3	20	Solution	30.01	776	754.00	2.84
260 50 3 20 rep 10.txt	50	3	20	Solution	30.01	751	735.00	2.13
261 50 3 30 rep 1.txt	50	3	30	Solution	30.01	548	538.00	1.82
262 50 3 30 rep 2.txt	50	3	30	Solution	30.01	648	635.00	2.01
263 50 3 30 rep 3.txt	50	3	30	Solution	30.01	521	500.00	4.03
264 50 3 30 rep 4.txt	50	3	30	Solution	30.01	525	501.00	4.57
265 50 3 30 rep 5.txt	50	3	30	Solution	30.01	578	562.00	2.77
266 50 3 30 rep 6.txt	50	3	30	Solution	30.01	546	529.00	3.11
267 50 3 30 rep 7.txt	50	3	30	Solution	30.01	616	598.00	2.92

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
268 50 3 30 rep 8.txt	50	3	30	Solution	30.01	549	535.00	2.55
269 50 3 30 rep 9.txt	50	3	30	Solution	30.01	606	589.00	2.81
270 50 3 30 rep 10.txt	50	3	30	Solution	30.01	584	572.00	2.05
271 50 3 50 rep 1.txt	50	3	50	Solution	30.01	352	343.00	2.56
272 50 3 50 rep 2.txt	50	3	50	Solution	30.01	363	355.00	2.20
273 50 3 50 rep 3.txt	50	3	50	Solution	30.01	344	333.00	3.20
274 50 3 50 rep 4.txt	50	3	50	Solution	30.01	378	372.00	1.59
275 50 3 50 rep 5.txt	50	3	50	Solution	30.01	354	338.00	4.52
276 50 3 50 rep 6.txt	50	3	50	Solution	30.01	377	360.00	4.51
277 50 3 50 rep 7.txt	50	3	50	Solution	30.01	377	366.00	2.92
278 50 3 50 rep 8.txt	50	3	50	Solution	30.01	329	326.00	0.91
279 50 3 50 rep 9.txt	50	3	50	Solution	30.01	373	358.00	4.02
280 50 3 50 rep 10.txt	50	3	50	Solution	30.01	357	350.00	1.96
281 50 5 10 rep 1.txt	50	5	10	Solution	30.02	1728	1657.00	4.11
282 50 5 10 rep 2.txt	50	5	10	Solution	30.02	1956	1846.00	5.62
283 50 5 10 rep 3.txt	50	5	10	Solution	30.01	1694	1525.00	9.98
284 50 5 10 rep 4.txt	50	5	10	Solution	30.01	1956	1742.00	10.94
285 50 5 10 rep 5.txt	50	5	10	Solution	30.02	1980	1770.00	10.61
286 50 5 10 rep 6.txt	50	5	10	Solution	30.01	1892	1823.00	3.65
287 50 5 10 rep 7.txt	50	5	10	Solution	30.02	1906	1724.00	9.55
288 50 5 10 rep 8.txt	50	5	10	Solution	30.01	1712	1545.00	9.75
289 50 5 10 rep 9.txt	50	5	10	Solution	30.02	1972	1771.00	10.19
290 50 5 10 rep 10.txt	50	5	10	Solution	30.02	1869	1697.00	9.20
291 50 5 20 rep 1.txt	50	5	20	Solution	30.02	932	912.00	2.15
292 50 5 20 rep 2.txt	50	5	20	Solution	30.01	931	853.00	8.38
293 50 5 20 rep 3.txt	50	5	20	Solution	30.01	901	857.00	4.88
294 50 5 20 rep 4.txt	50	5	20	Solution	30.01	1005	970.00	3.48
295 50 5 20 rep 5.txt	50	5	20	Solution	30.01	1040	963.00	7.40
296 50 5 20 rep 6.txt	50	5	20	Solution	30.01	930	887.00	4.62
297 50 5 20 rep 7.txt	50	5	20	Solution	30.01	913	881.00	3.50
298 50 5 20 rep 8.txt	50	5	20	Solution	30.01	922	847.00	8.13
299 50 5 20 rep 9.txt	50	5	20	Solution	30.01	882	850.00	3.63
300 50 5 20 rep 10.txt	50	5	20	Solution	30.01	898	869.00	3.23
301 50 5 30 rep 1.txt	50	5	30	Solution	30.01	649	603.00	7.09
302 50 5 30 rep 2.txt	50	5	30	Solution	30.02	647	595.00	8.04
303 50 5 30 rep 3.txt	50	5	30	Solution	30.01	638	587.00	7.99
304 50 5 30 rep 4.txt	50	5	30	Solution	30.01	659	611.00	7.28
305 50 5 30 rep 5.txt	50	5	30	Solution	30.01	675	628.00	6.96
306 50 5 30 rep 6.txt	50	5	30	Solution	30.01	696	655.00	5.89
307 50 5 30 rep 7.txt	50	5	30	Solution	30.01	675	621.00	8.00
308 50 5 30 rep 8.txt	50	5	30	Solution	30.01	737	706.00	4.21
309 50 5 30 rep 9.txt	50	5	30	Solution	30.01	652	606.00	7.06
310 50 5 30 rep 10.txt	50	5	30	Solution	30.01	711	678.00	4.64
311 50 5 50 rep 1.txt	50	5	50	Solution	30.01	427	401.00	6.09
312 50 5 50 rep 2.txt	50	5	50	Solution	30.01	471	459.00	2.55

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
313 50 5 50 rep 3.txt	50	5	50	Solution	30.02	466	445.00	4.51
314 50 5 50 rep 4.txt	50	5	50	Solution	30.01	458	419.00	8.52
315 50 5 50 rep 5.txt	50	5	50	Solution	30.01	463	439.00	5.18
316 50 5 50 rep 6.txt	50	5	50	Solution	30.01	419	398.00	5.01
317 50 5 50 rep 7.txt	50	5	50	Optimal	0.63	440	440.00	0.00
318 50 5 50 rep 8.txt	50	5	50	Solution	30.01	440	409.00	7.05
319 50 5 50 rep 9.txt	50	5	50	Solution	30.01	425	403.00	5.18
320 50 5 50 rep 10.txt	50	5	50	Solution	30.01	425	393.00	7.53
321 50 10 10 rep 1.txt	50	10	10	Solution	30.03	2226	1911.00	14.15
322 50 10 10 rep 2.txt	50	10	10	Solution	30.02	2215	1757.00	20.68
323 50 10 10 rep 3.txt	50	10	10	Solution	30.05	2486	2143.00	13.80
324 50 10 10 rep 4.txt	50	10	10	Solution	30.02	2173	1797.00	17.30
325 50 10 10 rep 5.txt	50	10	10	Solution	30.03	2366	1835.00	22.44
326 50 10 10 rep 6.txt	50	10	10	Solution	30.03	2392	1905.00	20.36
327 50 10 10 rep 7.txt	50	10	10	Solution	30.03	2388	2088.00	12.56
328 50 10 10 rep 8.txt	50	10	10	Solution	30.02	2334	2059.00	11.78
329 50 10 10 rep 9.txt	50	10	10	Solution	30.02	2151	1745.00	18.87
330 50 10 10 rep 10.txt	50	10	10	Solution	30.02	2395	1978.00	17.41
331 50 10 20 rep 1.txt	50	10	20	Solution	30.02	1236	1084.00	12.30
332 50 10 20 rep 2.txt	50	10	20	Solution	30.02	1225	1106.00	9.71
333 50 10 20 rep 3.txt	50	10	20	Solution	30.02	1317	1179.00	10.48
334 50 10 20 rep 4.txt	50	10	20	Solution	30.02	1239	1106.00	10.73
335 50 10 20 rep 5.txt	50	10	20	Solution	30.01	1182	1109.00	6.18
336 50 10 20 rep 6.txt	50	10	20	Solution	30.02	1234	1154.00	6.48
337 50 10 20 rep 7.txt	50	10	20	Solution	30.02	1211	1121.00	7.43
338 50 10 20 rep 8.txt	50	10	20	Solution	30.02	1192	1054.00	11.58
339 50 10 20 rep 9.txt	50	10	20	Solution	30.02	1383	1235.00	10.70
340 50 10 20 rep 10.txt	50	10	20	Solution	30.02	1093	950.00	13.08
341 50 10 30 rep 1.txt	50	10	30	Solution	30.02	931	851.00	8.59
342 50 10 30 rep 2.txt	50	10	30	Solution	30.02	935	816.00	12.73
343 50 10 30 rep 3.txt	50	10	30	Solution	30.02	978	864.00	11.66
344 50 10 30 rep 4.txt	50	10	30	Solution	30.01	935	827.00	11.55
345 50 10 30 rep 5.txt	50	10	30	Solution	30.02	954	860.00	9.85
346 50 10 30 rep 6.txt	50	10	30	Solution	30.02	1003	876.00	12.66
347 50 10 30 rep 7.txt	50	10	30	Solution	30.02	941	820.00	12.86
348 50 10 30 rep 8.txt	50	10	30	Solution	30.02	943	842.00	10.71
349 50 10 30 rep 9.txt	50	10	30	Solution	30.02	869	759.00	12.66
350 50 10 30 rep 10.txt	50	10	30	Solution	30.02	957	890.00	7.00
351 50 10 50 rep 1.txt	50	10	50	Solution	30.02	738	708.00	4.07
352 50 10 50 rep 2.txt	50	10	50	Solution	30.01	684	663.00	3.07
353 50 10 50 rep 3.txt	50	10	50	Solution	30.02	751	705.00	6.13
354 50 10 50 rep 4.txt	50	10	50	Optimal	3.26	748	748.00	0.00
355 50 10 50 rep 5.txt	50	10	50	Optimal	1.00	791	791.00	0.00
356 50 10 50 rep 6.txt	50	10	50	Solution	30.01	728	691.00	5.08
357 50 10 50 rep 7.txt	50	10	50	Solution	30.02	735	690.00	6.12

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
358 50 10 50 rep 8.txt	50	10	50	Optimal	19.73	713	713.00	0.00
359 50 10 50 rep 9.txt	50	10	50	Solution	30.01	745	702.00	5.77
360 50 10 50 rep 10.txt	50	10	50	Solution	30.02	732	691.00	5.60
361 100 3 10 rep 1.txt	100	3	10	Solution	30.03	3199	2997.00	6.31
362 100 3 10 rep 2.txt	100	3	10	Solution	30.03	3414	3166.00	7.26
363 100 3 10 rep 3.txt	100	3	10	Solution	30.02	3123	2973.00	4.80
364 100 3 10 rep 4.txt	100	3	10	Solution	30.02	3245	3035.00	6.47
365 100 3 10 rep 5.txt	100	3	10	Solution	30.02	3403	3166.00	6.96
366 100 3 10 rep 6.txt	100	3	10	Solution	30.02	3482	3237.00	7.04
367 100 3 10 rep 7.txt	100	3	10	Solution	30.03	3459	3212.00	7.14
368 100 3 10 rep 8.txt	100	3	10	Solution	30.02	3482	3275.00	5.94
369 100 3 10 rep 9.txt	100	3	10	Solution	30.02	3228	3013.00	6.66
370 100 3 10 rep 10.txt	100	3	10	Solution	30.02	3271	3065.00	6.30
371 100 3 20 rep 1.txt	100	3	20	Solution	30.02	1570	1516.00	3.44
372 100 3 20 rep 2.txt	100	3	20	Solution	30.02	1419	1381.00	2.68
373 100 3 20 rep 3.txt	100	3	20	Solution	30.02	1544	1489.00	3.56
374 100 3 20 rep 4.txt	100	3	20	Solution	30.01	1444	1405.00	2.70
375 100 3 20 rep 5.txt	100	3	20	Solution	30.02	1694	1657.00	2.18
376 100 3 20 rep 6.txt	100	3	20	Solution	30.02	1652	1601.00	3.09
377 100 3 20 rep 7.txt	100	3	20	Solution	30.02	1622	1578.00	2.71
378 100 3 20 rep 8.txt	100	3	20	Solution	30.02	1516	1478.00	2.51
379 100 3 20 rep 9.txt	100	3	20	Solution	30.02	1612	1582.00	1.86
380 100 3 20 rep 10.txt	100	3	20	Solution	30.02	1436	1397.00	2.72
381 100 3 30 rep 1.txt	100	3	30	Solution	30.02	1029	999.00	2.92
382 100 3 30 rep 2.txt	100	3	30	Solution	30.02	1098	1070.00	2.55
383 100 3 30 rep 3.txt	100	3	30	Solution	30.02	1035	1019.00	1.55
384 100 3 30 rep 4.txt	100	3	30	Solution	30.01	1130	1100.00	2.65
385 100 3 30 rep 5.txt	100	3	30	Solution	30.02	1247	1225.00	1.76
386 100 3 30 rep 6.txt	100	3	30	Solution	30.02	976	964.00	1.23
387 100 3 30 rep 7.txt	100	3	30	Solution	30.01	1089	1057.00	2.94
388 100 3 30 rep 8.txt	100	3	30	Solution	30.02	1070	1043.00	2.52
389 100 3 30 rep 9.txt	100	3	30	Solution	30.01	1083	1067.00	1.48
390 100 3 30 rep 10.txt	100	3	30	Solution	30.01	1100	1086.00	1.27
391 100 3 50 rep 1.txt	100	3	50	Solution	30.02	561	545.00	2.85
392 100 3 50 rep 2.txt	100	3	50	Solution	30.01	620	611.00	1.45
393 100 3 50 rep 3.txt	100	3	50	Solution	30.01	638	621.00	2.66
394 100 3 50 rep 4.txt	100	3	50	Solution	30.02	720	709.00	1.53
395 100 3 50 rep 5.txt	100	3	50	Solution	30.01	626	604.00	3.51
396 100 3 50 rep 6.txt	100	3	50	Solution	30.01	661	649.00	1.82
397 100 3 50 rep 7.txt	100	3	50	Solution	30.01	656	636.00	3.05
398 100 3 50 rep 8.txt	100	3	50	Solution	30.02	651	643.00	1.23
399 100 3 50 rep 9.txt	100	3	50	Solution	30.01	612	606.00	0.98
400 100 3 50 rep 10.txt	100	3	50	Solution	30.01	638	613.00	3.92
401 100 5 10 rep 1.txt	100	5	10	Solution	30.04	3300	2911.00	11.79
402 100 5 10 rep 2.txt	100	5	10	Solution	30.03	3500	3283.00	6.20

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
403 100 5 10 rep 3.txt	100	5	10	Solution	30.04	3811	3371.00	11.55
404 100 5 10 rep 4.txt	100	5	10	Solution	30.03	3409	3109.00	8.80
405 100 5 10 rep 5.txt	100	5	10	Solution	30.04	3606	3203.00	11.18
406 100 5 10 rep 6.txt	100	5	10	Solution	30.04	3471	3250.00	6.37
407 100 5 10 rep 7.txt	100	5	10	Solution	30.04	3372	3105.00	7.92
408 100 5 10 rep 8.txt	100	5	10	Solution	30.03	3481	3240.00	6.92
409 100 5 10 rep 9.txt	100	5	10	Solution	30.03	3285	2945.00	10.35
410 100 5 10 rep 10.txt	100	5	10	Solution	30.04	3562	3382.00	5.05
411 100 5 20 rep 1.txt	100	5	20	Solution	30.03	1883	1820.00	3.35
412 100 5 20 rep 2.txt	100	5	20	Solution	30.02	1651	1555.00	5.81
413 100 5 20 rep 3.txt	100	5	20	Solution	30.02	1722	1645.00	4.47
414 100 5 20 rep 4.txt	100	5	20	Solution	30.02	1669	1613.00	3.36
415 100 5 20 rep 5.txt	100	5	20	Solution	30.02	1724	1642.00	4.76
416 100 5 20 rep 6.txt	100	5	20	Solution	30.03	1634	1542.00	5.63
417 100 5 20 rep 7.txt	100	5	20	Solution	30.03	1859	1796.00	3.39
418 100 5 20 rep 8.txt	100	5	20	Solution	30.02	1589	1505.00	5.29
419 100 5 20 rep 9.txt	100	5	20	Solution	30.02	1561	1408.00	9.80
420 100 5 20 rep 10.txt	100	5	20	Solution	30.02	1781	1702.00	4.44
421 100 5 30 rep 1.txt	100	5	30	Solution	30.01	1156	1095.00	5.28
422 100 5 30 rep 2.txt	100	5	30	Solution	30.04	1157	1081.00	6.57
423 100 5 30 rep 3.txt	100	5	30	Solution	30.02	1202	1168.00	2.83
424 100 5 30 rep 4.txt	100	5	30	Solution	30.03	1126	1063.00	5.60
425 100 5 30 rep 5.txt	100	5	30	Solution	30.02	1133	1080.00	4.68
426 100 5 30 rep 6.txt	100	5	30	Solution	30.02	1161	1111.00	4.31
427 100 5 30 rep 7.txt	100	5	30	Solution	30.03	1175	1101.00	6.30
428 100 5 30 rep 8.txt	100	5	30	Solution	30.02	1194	1144.00	4.19
429 100 5 30 rep 9.txt	100	5	30	Solution	30.02	1208	1155.00	4.39
430 100 5 30 rep 10.txt	100	5	30	Solution	30.03	1139	1044.00	8.34
431 100 5 50 rep 1.txt	100	5	50	Solution	30.02	719	703.00	2.23
432 100 5 50 rep 2.txt	100	5	50	Solution	30.01	672	636.00	5.36
433 100 5 50 rep 3.txt	100	5	50	Solution	30.02	692	661.00	4.48
434 100 5 50 rep 4.txt	100	5	50	Solution	30.02	722	680.00	5.82
435 100 5 50 rep 5.txt	100	5	50	Solution	30.02	771	753.00	2.33
436 100 5 50 rep 6.txt	100	5	50	Solution	30.02	678	650.00	4.13
437 100 5 50 rep 7.txt	100	5	50	Solution	30.02	738	682.00	7.59
438 100 5 50 rep 8.txt	100	5	50	Solution	30.03	646	597.00	7.59
439 100 5 50 rep 9.txt	100	5	50	Solution	30.03	705	675.00	4.26
440 100 5 50 rep 10.txt	100	5	50	Solution	30.02	731	688.00	5.88
441 100 10 10 rep 1.txt	100	10	10	Solution	30.09	4041	3377.00	16.43
442 100 10 10 rep 2.txt	100	10	10	Solution	30.08	4042	3150.00	22.07
443 100 10 10 rep 3.txt	100	10	10	Solution	30.09	4231	3281.00	22.45
444 100 10 10 rep 4.txt	100	10	10	Solution	30.08	4270	3654.00	14.43
445 100 10 10 rep 5.txt	100	10	10	Solution	30.10	4314	3447.00	20.10
446 100 10 10 rep 6.txt	100	10	10	Solution	30.08	4024	3217.00	20.05
447 100 10 10 rep 7.txt	100	10	10	Solution	30.08	4214	3485.00	17.30

Table 1: Results for CFSprec (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
448 100 10 10 rep 8.txt	100	10	10	Solution	30.09	4341	3527.00	18.75
449 100 10 10 rep 9.txt	100	10	10	Solution	30.09	4282	3435.00	19.78
450 100 10 10 rep 10.txt	100	10	10	Solution	30.09	4430	3851.00	13.07
451 100 10 20 rep 1.txt	100	10	20	Solution	30.05	2100	1897.00	9.67
452 100 10 20 rep 2.txt	100	10	20	Solution	30.06	2145	1825.00	14.92
453 100 10 20 rep 3.txt	100	10	20	Solution	30.10	2428	2209.00	9.02
454 100 10 20 rep 4.txt	100	10	20	Solution	30.04	2118	1797.00	15.16
455 100 10 20 rep 5.txt	100	10	20	Solution	30.06	2271	1971.00	13.21
456 100 10 20 rep 6.txt	100	10	20	Solution	30.05	1975	1656.00	16.15
457 100 10 20 rep 7.txt	100	10	20	Solution	30.04	2144	1746.00	18.56
458 100 10 20 rep 8.txt	100	10	20	Solution	30.06	2158	1958.00	9.27
459 100 10 20 rep 9.txt	100	10	20	Solution	30.06	2048	1834.00	10.45
460 100 10 20 rep 10.txt	100	10	20	Solution	30.05	2103	1786.00	15.07
461 100 10 30 rep 1.txt	100	10	30	Solution	30.04	1534	1283.00	16.36
462 100 10 30 rep 2.txt	100	10	30	Solution	30.03	1475	1258.00	14.71
463 100 10 30 rep 3.txt	100	10	30	Solution	30.04	1581	1425.00	9.87
464 100 10 30 rep 4.txt	100	10	30	Solution	30.04	1501	1267.00	15.59
465 100 10 30 rep 5.txt	100	10	30	Solution	30.05	1493	1294.00	13.33
466 100 10 30 rep 6.txt	100	10	30	Solution	30.05	1539	1283.00	16.63
467 100 10 30 rep 7.txt	100	10	30	Solution	30.04	1522	1357.00	10.84
468 100 10 30 rep 8.txt	100	10	30	Solution	30.03	1496	1321.00	11.70
469 100 10 30 rep 9.txt	100	10	30	Solution	30.04	1512	1342.00	11.24
470 100 10 30 rep 10.txt	100	10	30	Solution	30.04	1520	1297.00	14.67
471 100 10 50 rep 1.txt	100	10	50	Solution	30.04	999	923.00	7.61
472 100 10 50 rep 2.txt	100	10	50	Solution	30.04	1001	900.00	10.09
473 100 10 50 rep 3.txt	100	10	50	Solution	30.05	1056	914.00	13.45
474 100 10 50 rep 4.txt	100	10	50	Solution	30.06	1009	911.00	9.71
475 100 10 50 rep 5.txt	100	10	50	Solution	30.04	1011	893.00	11.67
476 100 10 50 rep 6.txt	100	10	50	Solution	30.04	1043	947.00	9.20
477 100 10 50 rep 7.txt	100	10	50	Solution	30.02	1032	912.00	11.63
478 100 10 50 rep 8.txt	100	10	50	Solution	30.04	1021	895.00	12.34
479 100 10 50 rep 9.txt	100	10	50	Solution	30.04	1001	868.00	13.29
480 100 10 50 rep 10.txt	100	10	50	Solution	30.04	1013	898.00	11.35

2 Instances Precedence (CPSat)

Table 2: Results for CFSprec (CPSat) (245 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
1 20 3 10 rep 1.txt	20	3	10	Optimal	4.92	741	741.00	0.00
2 20 3 10 rep 2.txt	20	3	10	Solution	30.14	715	691.00	3.36

Table 2: Results for CFSprec (CPSat) (245 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
3 20 3 10 rep 3.txt	20	3	10	Solution	30.02	491	477.00	2.85
4 20 3 10 rep 4.txt	20	3	10	Optimal	30.01	742	742.00	0.00
5 20 3 10 rep 5.txt	20	3	10	Solution	30.02	880	871.00	1.02
6 20 3 10 rep 6.txt	20	3	10	Optimal	30.01	670	670.00	0.00
7 20 3 10 rep 7.txt	20	3	10	Solution	30.04	715	704.00	1.54
8 20 3 10 rep 8.txt	20	3	10	Optimal	5.58	813	813.00	0.00
9 20 3 10 rep 9.txt	20	3	10	Solution	30.02	581	559.00	3.79
10 20 3 10 rep 10.txt	20	3	10	Optimal	30.01	784	784.00	0.00
11 20 3 20 rep 1.txt	20	3	20	Optimal	30.01	400	400.00	0.00
12 20 3 20 rep 2.txt	20	3	20	Optimal	30.00	355	355.00	0.00
13 20 3 20 rep 3.txt	20	3	20	Solution	30.05	421	414.00	1.66
14 20 3 20 rep 4.txt	20	3	20	Solution	30.02	376	343.00	8.78
15 20 3 20 rep 5.txt	20	3	20	Optimal	30.01	344	344.00	0.00
16 20 3 20 rep 6.txt	20	3	20	Optimal	30.01	377	377.00	0.00
17 20 3 20 rep 7.txt	20	3	20	Solution	30.02	369	345.00	6.50
18 20 3 20 rep 8.txt	20	3	20	Optimal	30.01	412	412.00	0.00
19 20 3 20 rep 9.txt	20	3	20	Solution	30.03	294	280.00	4.76
20 20 3 20 rep 10.txt	20	3	20	Optimal	30.00	349	349.00	0.00
21 20 3 30 rep 1.txt	20	3	30	Optimal	5.73	335	335.00	0.00
22 20 3 30 rep 2.txt	20	3	30	Optimal	30.00	295	295.00	0.00
23 20 3 30 rep 3.txt	20	3	30	Optimal	30.01	301	301.00	0.00
24 20 3 30 rep 4.txt	20	3	30	Optimal	30.01	289	289.00	0.00
25 20 3 30 rep 5.txt	20	3	30	Optimal	30.01	292	292.00	0.00
26 20 3 30 rep 6.txt	20	3	30	Optimal	13.12	287	287.00	0.00
27 20 3 30 rep 7.txt	20	3	30	Optimal	30.00	345	345.00	0.00
28 20 3 30 rep 8.txt	20	3	30	Optimal	3.14	322	322.00	0.00
29 20 3 30 rep 9.txt	20	3	30	Optimal	6.85	300	300.00	0.00
30 20 3 30 rep 10.txt	20	3	30	Optimal	30.01	245	245.00	0.00
31 20 3 50 rep 1.txt	20	3	50	Optimal	0.01	253	253.00	0.00
32 20 3 50 rep 2.txt	20	3	50	Optimal	30.01	237	237.00	0.00
33 20 3 50 rep 3.txt	20	3	50	Optimal	0.02	241	241.00	0.00
34 20 3 50 rep 4.txt	20	3	50	Optimal	0.08	231	231.00	0.00
35 20 3 50 rep 5.txt	20	3	50	Optimal	0.02	272	272.00	0.00
36 20 3 50 rep 6.txt	20	3	50	Optimal	0.01	216	216.00	0.00
37 20 3 50 rep 7.txt	20	3	50	Optimal	0.04	224	224.00	0.00
38 20 3 50 rep 8.txt	20	3	50	Optimal	0.01	264	264.00	0.00
39 20 3 50 rep 9.txt	20	3	50	Optimal	0.13	234	234.00	0.00
40 20 3 50 rep 10.txt	20	3	50	Optimal	1.14	231	231.00	0.00
41 20 5 10 rep 1.txt	20	5	10	Solution	30.02	898	861.00	4.12
42 20 5 10 rep 2.txt	20	5	10	Optimal	30.01	869	869.00	0.00
43 20 5 10 rep 3.txt	20	5	10	Solution	30.02	861	851.00	1.16
44 20 5 10 rep 4.txt	20	5	10	Solution	30.02	843	804.00	4.63
45 20 5 10 rep 5.txt	20	5	10	Solution	30.04	849	835.00	1.65
46 20 5 10 rep 6.txt	20	5	10	Optimal	30.00	814	814.00	0.00
47 20 5 10 rep 7.txt	20	5	10	Solution	30.02	853	793.00	7.03

Table 2: Results for CFSprec (CPSat) (245 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
48 20 5 10 rep 8.txt	20	5	10	Solution	30.02	811	802.00	1.11
49 20 5 10 rep 9.txt	20	5	10	Optimal	5.73	844	844.00	0.00
50 20 5 10 rep 10.txt	20	5	10	Optimal	30.01	1068	1068.00	0.00
51 20 5 20 rep 1.txt	20	5	20	Solution	30.02	490	469.00	4.29
52 20 5 20 rep 2.txt	20	5	20	Optimal	30.01	434	434.00	0.00
53 20 5 20 rep 3.txt	20	5	20	Solution	30.02	514	481.00	6.42
54 20 5 20 rep 4.txt	20	5	20	Solution	30.02	387	368.00	4.91
55 20 5 20 rep 5.txt	20	5	20	Optimal	30.00	470	470.00	0.00
56 20 5 20 rep 6.txt	20	5	20	Solution	30.08	501	470.00	6.19
57 20 5 20 rep 7.txt	20	5	20	Optimal	30.00	459	459.00	0.00
58 20 5 20 rep 8.txt	20	5	20	Optimal	30.00	496	496.00	0.00
59 20 5 20 rep 9.txt	20	5	20	Optimal	30.00	518	518.00	0.00
60 20 5 20 rep 10.txt	20	5	20	Solution	30.02	479	430.00	10.23
61 20 5 30 rep 1.txt	20	5	30	Optimal	9.19	386	386.00	0.00
62 20 5 30 rep 2.txt	20	5	30	Optimal	30.00	451	451.00	0.00
63 20 5 30 rep 3.txt	20	5	30	Optimal	0.26	367	367.00	0.00
64 20 5 30 rep 4.txt	20	5	30	Solution	30.02	422	408.00	3.32
65 20 5 30 rep 5.txt	20	5	30	Optimal	30.01	380	380.00	0.00
66 20 5 30 rep 6.txt	20	5	30	Optimal	30.01	401	401.00	0.00
67 20 5 30 rep 7.txt	20	5	30	Optimal	0.02	431	431.00	0.00
68 20 5 30 rep 8.txt	20	5	30	Optimal	30.01	404	404.00	0.00
69 20 5 30 rep 9.txt	20	5	30	Optimal	14.00	435	435.00	0.00
70 20 5 30 rep 10.txt	20	5	30	Optimal	30.01	436	436.00	0.00
71 20 5 50 rep 1.txt	20	5	50	Optimal	0.03	312	312.00	0.00
72 20 5 50 rep 2.txt	20	5	50	Optimal	1.78	323	323.00	0.00
73 20 5 50 rep 3.txt	20	5	50	Optimal	0.02	353	353.00	0.00
74 20 5 50 rep 4.txt	20	5	50	Optimal	0.02	348	348.00	0.00
75 20 5 50 rep 5.txt	20	5	50	Optimal	0.02	371	371.00	0.00
76 20 5 50 rep 6.txt	20	5	50	Optimal	0.02	425	425.00	0.00
77 20 5 50 rep 7.txt	20	5	50	Optimal	0.01	349	349.00	0.00
78 20 5 50 rep 8.txt	20	5	50	Optimal	0.02	398	398.00	0.00
79 20 5 50 rep 9.txt	20	5	50	Optimal	0.01	398	398.00	0.00
80 20 5 50 rep 10.txt	20	5	50	Optimal	0.03	386	386.00	0.00
81 20 10 10 rep 1.txt	20	10	10	Solution	30.02	1222	1026.00	16.04
82 20 10 10 rep 2.txt	20	10	10	Solution	30.02	1231	1031.00	16.25
83 20 10 10 rep 3.txt	20	10	10	Solution	30.02	1252	1139.00	9.03
84 20 10 10 rep 4.txt	20	10	10	Solution	30.02	1226	1168.00	4.73
85 20 10 10 rep 5.txt	20	10	10	Solution	30.02	1300	1162.00	10.62
86 20 10 10 rep 6.txt	20	10	10	Solution	30.02	1230	1169.00	4.96
87 20 10 10 rep 7.txt	20	10	10	Solution	30.04	1195	1046.00	12.47
88 20 10 10 rep 8.txt	20	10	10	Solution	30.02	1249	1197.00	4.16
89 20 10 10 rep 9.txt	20	10	10	Solution	30.03	1222	1081.00	11.54
90 20 10 10 rep 10.txt	20	10	10	Solution	30.03	1211	1144.00	5.53
91 20 10 20 rep 1.txt	20	10	20	Solution	30.02	739	694.00	6.09
92 20 10 20 rep 2.txt	20	10	20	Solution	30.02	829	733.00	11.58

Table 2: Results for CFSprec (CPSat) (245 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
93 20 10 20 rep 3.txt	20	10	20	Optimal	30.01	699	699.00	0.00
94 20 10 20 rep 4.txt	20	10	20	Solution	30.02	836	752.00	10.05
95 20 10 20 rep 5.txt	20	10	20	Solution	30.02	722	695.00	3.74
96 20 10 20 rep 6.txt	20	10	20	Solution	30.02	873	802.00	8.13
97 20 10 20 rep 7.txt	20	10	20	Solution	30.02	792	730.00	7.83
98 20 10 20 rep 8.txt	20	10	20	Solution	30.02	803	749.00	6.72
99 20 10 20 rep 9.txt	20	10	20	Solution	30.02	797	718.00	9.91
100 20 10 20 rep 10.txt	20	10	20	Solution	30.02	745	724.00	2.82
101 20 10 30 rep 1.txt	20	10	30	Optimal	30.01	730	730.00	0.00
102 20 10 30 rep 2.txt	20	10	30	Optimal	0.17	780	780.00	0.00
103 20 10 30 rep 3.txt	20	10	30	Optimal	2.39	614	614.00	0.00
104 20 10 30 rep 4.txt	20	10	30	Optimal	1.28	726	726.00	0.00
105 20 10 30 rep 5.txt	20	10	30	Solution	30.02	670	655.00	2.24
106 20 10 30 rep 6.txt	20	10	30	Optimal	1.14	639	639.00	0.00
107 20 10 30 rep 7.txt	20	10	30	Optimal	17.03	728	728.00	0.00
108 20 10 30 rep 8.txt	20	10	30	Optimal	0.22	763	763.00	0.00
109 20 10 30 rep 9.txt	20	10	30	Optimal	2.38	645	645.00	0.00
110 20 10 30 rep 10.txt	20	10	30	Optimal	0.13	692	692.00	0.00
111 20 10 50 rep 1.txt	20	10	50	Optimal	0.03	668	668.00	0.00
112 20 10 50 rep 2.txt	20	10	50	Optimal	0.03	740	740.00	0.00
113 20 10 50 rep 3.txt	20	10	50	Optimal	0.03	701	701.00	0.00
114 20 10 50 rep 4.txt	20	10	50	Optimal	0.04	693	693.00	0.00
115 20 10 50 rep 5.txt	20	10	50	Optimal	0.04	651	651.00	0.00
116 20 10 50 rep 6.txt	20	10	50	Optimal	0.05	641	641.00	0.00
117 20 10 50 rep 7.txt	20	10	50	Optimal	0.04	654	654.00	0.00
118 20 10 50 rep 8.txt	20	10	50	Optimal	0.04	673	673.00	0.00
119 20 10 50 rep 9.txt	20	10	50	Optimal	0.02	708	708.00	0.00
120 20 10 50 rep 10.txt	20	10	50	Optimal	0.02	731	731.00	0.00
121 30 3 10 rep 1.txt	30	3	10	Solution	30.02	931	897.00	3.65
122 30 3 10 rep 2.txt	30	3	10	Solution	30.02	1242	1237.00	0.40
123 30 3 10 rep 3.txt	30	3	10	Solution	30.05	932	907.00	2.68
124 30 3 10 rep 4.txt	30	3	10	Solution	30.02	1208	1199.00	0.75
125 30 3 10 rep 5.txt	30	3	10	Optimal	30.01	1105	1105.00	0.00
126 30 3 10 rep 6.txt	30	3	10	Solution	30.02	1200	1198.00	0.17
127 30 3 10 rep 7.txt	30	3	10	Optimal	30.00	1221	1221.00	0.00
128 30 3 10 rep 8.txt	30	3	10	Optimal	30.01	1181	1181.00	0.00
129 30 3 10 rep 9.txt	30	3	10	Solution	30.02	1019	972.00	4.61
130 30 3 10 rep 10.txt	30	3	10	Optimal	30.01	1039	1039.00	0.00
131 30 3 20 rep 1.txt	30	3	20	Solution	30.02	548	527.00	3.83
132 30 3 20 rep 2.txt	30	3	20	Solution	30.02	566	541.00	4.42
133 30 3 20 rep 3.txt	30	3	20	Solution	30.01	473	439.00	7.19
134 30 3 20 rep 4.txt	30	3	20	Solution	30.03	618	597.00	3.40
135 30 3 20 rep 5.txt	30	3	20	Solution	30.02	450	423.00	6.00
136 30 3 20 rep 6.txt	30	3	20	Solution	30.02	494	488.00	1.21
137 30 3 20 rep 7.txt	30	3	20	Solution	30.03	503	474.00	5.77

Table 2: Results for CFSprec (CPSat) (245 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
138 30 3 20 rep 8.txt	30	3	20	Solution	30.02	545	537.00	1.47
139 30 3 20 rep 9.txt	30	3	20	Solution	30.02	498	486.00	2.41
140 30 3 20 rep 10.txt	30	3	20	Solution	30.02	468	453.00	3.21
141 30 3 30 rep 1.txt	30	3	30	Solution	30.02	437	425.00	2.75
142 30 3 30 rep 2.txt	30	3	30	Solution	30.01	328	317.00	3.35
143 30 3 30 rep 3.txt	30	3	30	Solution	30.02	447	439.00	1.79
144 30 3 30 rep 4.txt	30	3	30	Solution	30.02	337	333.00	1.19
145 30 3 30 rep 5.txt	30	3	30	Solution	30.02	430	423.00	1.63
146 30 3 30 rep 6.txt	30	3	30	Solution	30.01	325	303.00	6.77
147 30 3 30 rep 7.txt	30	3	30	Solution	30.02	416	407.00	2.16
148 30 3 30 rep 8.txt	30	3	30	Solution	30.03	393	382.00	2.80
149 30 3 30 rep 9.txt	30	3	30	Solution	30.03	414	412.00	0.48
150 30 3 30 rep 10.txt	30	3	30	Solution	30.02	363	358.00	1.38
151 30 3 50 rep 1.txt	30	3	50	Optimal	30.01	252	252.00	0.00
152 30 3 50 rep 2.txt	30	3	50	Optimal	30.00	258	258.00	0.00
153 30 3 50 rep 3.txt	30	3	50	Optimal	0.24	294	294.00	0.00
154 30 3 50 rep 4.txt	30	3	50	Optimal	30.01	235	235.00	0.00
155 30 3 50 rep 5.txt	30	3	50	Solution	30.01	253	250.00	1.19
156 30 3 50 rep 6.txt	30	3	50	Optimal	0.81	268	268.00	0.00
157 30 3 50 rep 7.txt	30	3	50	Optimal	30.01	252	252.00	0.00
158 30 3 50 rep 8.txt	30	3	50	Solution	30.02	259	256.00	1.16
159 30 3 50 rep 9.txt	30	3	50	Optimal	30.01	275	275.00	0.00
160 30 3 50 rep 10.txt	30	3	50	Solution	30.02	264	258.00	2.27
161 30 5 10 rep 1.txt	30	5	10	Solution	30.02	1228	1197.00	2.52
162 30 5 10 rep 2.txt	30	5	10	Solution	30.02	1206	1175.00	2.57
163 30 5 10 rep 3.txt	30	5	10	Solution	30.02	1240	1156.00	6.77
164 30 5 10 rep 4.txt	30	5	10	Optimal	30.00	1119	1119.00	0.00
165 30 5 10 rep 5.txt	30	5	10	Solution	30.02	1334	1330.00	0.30
166 30 5 10 rep 6.txt	30	5	10	Solution	30.02	1121	1098.00	2.05
167 30 5 10 rep 7.txt	30	5	10	Optimal	30.00	1182	1182.00	0.00
168 30 5 10 rep 8.txt	30	5	10	Optimal	30.00	1541	1541.00	0.00
169 30 5 10 rep 9.txt	30	5	10	Optimal	30.01	1022	1022.00	0.00
170 30 5 10 rep 10.txt	30	5	10	Solution	30.03	1226	1193.00	2.69
171 30 5 20 rep 1.txt	30	5	20	Solution	30.02	666	581.00	12.76
172 30 5 20 rep 2.txt	30	5	20	Solution	30.03	607	584.00	3.79
173 30 5 20 rep 3.txt	30	5	20	Solution	30.03	657	616.00	6.24
174 30 5 20 rep 4.txt	30	5	20	Solution	30.02	677	641.00	5.32
175 30 5 20 rep 5.txt	30	5	20	Solution	30.06	565	531.00	6.02
176 30 5 20 rep 6.txt	30	5	20	Solution	30.04	678	654.00	3.54
177 30 5 20 rep 7.txt	30	5	20	Solution	30.02	664	604.00	9.04
178 30 5 20 rep 8.txt	30	5	20	Solution	30.03	627	548.00	12.60
179 30 5 20 rep 9.txt	30	5	20	Solution	30.02	611	536.00	12.27
180 30 5 20 rep 10.txt	30	5	20	Solution	30.02	669	654.00	2.24
181 30 5 30 rep 1.txt	30	5	30	Solution	30.02	454	431.00	5.07
182 30 5 30 rep 2.txt	30	5	30	Solution	30.02	474	446.00	5.91

Table 2: Results for CFSprec (CPSat) (245 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
183 30 5 30 rep 3.txt	30	5	30	Solution	30.01	482	455.00	5.60
184 30 5 30 rep 4.txt	30	5	30	Solution	30.01	481	452.00	6.03
185 30 5 30 rep 5.txt	30	5	30	Solution	30.01	505	468.00	7.33
186 30 5 30 rep 6.txt	30	5	30	Solution	30.02	410	397.00	3.17
187 30 5 30 rep 7.txt	30	5	30	Solution	30.02	471	447.00	5.10
188 30 5 30 rep 8.txt	30	5	30	Solution	30.01	471	460.00	2.34
189 30 5 30 rep 9.txt	30	5	30	Solution	30.01	472	449.00	4.87
190 30 5 30 rep 10.txt	30	5	30	Solution	30.01	515	467.00	9.32
191 30 5 50 rep 1.txt	30	5	50	Solution	30.02	368	366.00	0.54
192 30 5 50 rep 2.txt	30	5	50	Optimal	0.14	378	378.00	0.00
193 30 5 50 rep 3.txt	30	5	50	Optimal	30.00	378	378.00	0.00
194 30 5 50 rep 4.txt	30	5	50	Optimal	30.00	382	382.00	0.00
195 30 5 50 rep 5.txt	30	5	50	Optimal	30.00	365	365.00	0.00
196 30 5 50 rep 6.txt	30	5	50	Optimal	0.10	379	379.00	0.00
197 30 5 50 rep 7.txt	30	5	50	Optimal	30.00	394	394.00	0.00
198 30 5 50 rep 8.txt	30	5	50	Optimal	0.07	445	445.00	0.00
199 30 5 50 rep 9.txt	30	5	50	Optimal	0.87	360	360.00	0.00
200 30 5 50 rep 10.txt	30	5	50	Optimal	30.00	373	373.00	0.00
201 30 10 10 rep 1.txt	30	10	10	Solution	30.02	1748	1605.00	8.18
202 30 10 10 rep 2.txt	30	10	10	Solution	30.02	1524	1188.00	22.05
203 30 10 10 rep 3.txt	30	10	10	Solution	30.24	1582	1292.00	18.33
204 30 10 10 rep 4.txt	30	10	10	Solution	30.02	1619	1201.00	25.82
205 30 10 10 rep 5.txt	30	10	10	Solution	30.13	1577	1303.00	17.37
206 30 10 10 rep 6.txt	30	10	10	Solution	30.02	1686	1544.00	8.42
207 30 10 10 rep 7.txt	30	10	10	Solution	30.03	1571	1346.00	14.32
208 30 10 10 rep 8.txt	30	10	10	Solution	30.01	1607	1291.00	19.66
209 30 10 10 rep 9.txt	30	10	10	Solution	30.02	1655	1316.00	20.48
210 30 10 10 rep 10.txt	30	10	10	Solution	30.02	1674	1433.00	14.40
211 30 10 20 rep 1.txt	30	10	20	Solution	30.03	972	824.00	15.23
212 30 10 20 rep 2.txt	30	10	20	Solution	30.02	864	726.00	15.97
213 30 10 20 rep 3.txt	30	10	20	Solution	30.02	989	857.00	13.35
214 30 10 20 rep 4.txt	30	10	20	Solution	30.02	901	793.00	11.99
215 30 10 20 rep 5.txt	30	10	20	Solution	30.02	929	852.00	8.29
216 30 10 20 rep 6.txt	30	10	20	Solution	30.02	1011	893.00	11.67
217 30 10 20 rep 7.txt	30	10	20	Solution	30.02	989	867.00	12.34
218 30 10 20 rep 8.txt	30	10	20	Solution	30.07	940	865.00	7.98
219 30 10 20 rep 9.txt	30	10	20	Solution	30.02	909	787.00	13.42
220 30 10 20 rep 10.txt	30	10	20	Solution	30.01	949	811.00	14.54
221 30 10 30 rep 1.txt	30	10	30	Solution	30.01	760	720.00	5.26
222 30 10 30 rep 2.txt	30	10	30	Solution	30.02	780	742.00	4.87
223 30 10 30 rep 3.txt	30	10	30	Solution	30.02	699	698.00	0.14
224 30 10 30 rep 4.txt	30	10	30	Solution	30.02	722	710.00	1.66
225 30 10 30 rep 5.txt	30	10	30	Solution	30.02	762	698.00	8.40
226 30 10 30 rep 6.txt	30	10	30	Solution	30.05	787	730.00	7.24
227 30 10 30 rep 7.txt	30	10	30	Solution	30.01	736	692.00	5.98

Table 2: Results for CFSprec (CPSat) (245 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
228 30 10 30 rep 8.txt	30	10	30	Optimal	30.01	791	791.00	0.00
229 30 10 30 rep 9.txt	30	10	30	Solution	30.01	751	710.00	5.46
230 30 10 30 rep 10.txt	30	10	30	Solution	30.01	777	754.00	2.96
231 30 10 50 rep 1.txt	30	10	50	Optimal	0.07	738	738.00	0.00
232 30 10 50 rep 2.txt	30	10	50	Optimal	30.00	646	646.00	0.00
233 30 10 50 rep 3.txt	30	10	50	Optimal	0.09	753	753.00	0.00
234 30 10 50 rep 4.txt	30	10	50	Optimal	0.18	652	652.00	0.00
235 30 10 50 rep 5.txt	30	10	50	Optimal	30.01	667	667.00	0.00
236 30 10 50 rep 6.txt	30	10	50	Optimal	0.07	687	687.00	0.00
237 30 10 50 rep 7.txt	30	10	50	Optimal	0.08	827	827.00	0.00
238 30 10 50 rep 8.txt	30	10	50	Optimal	0.08	692	692.00	0.00
239 30 10 50 rep 9.txt	30	10	50	Optimal	0.14	671	671.00	0.00
240 30 10 50 rep 10.txt	30	10	50	Optimal	0.13	764	764.00	0.00
241 50 3 10 rep 1.txt	50	3	10	Solution	30.02	1787	1560.00	12.70
242 50 3 10 rep 2.txt	50	3	10	Solution	30.02	1682	1489.00	11.47
243 50 3 10 rep 3.txt	50	3	10	Solution	30.02	1718	1464.00	14.78
244 50 3 10 rep 4.txt	50	3	10	Solution	30.02	1834	1540.00	16.03
245 50 3 10 rep 5.txt	50	3	10	Solution	30.04	1597	1582.00	0.94

3 Comparison (Precedence)

Table 3: Comparison of CPO and CPSat for Results of CFS Prec

Group	Nr	All Instances Optimal (% of All Instances)				Optimal Only Time (% of VB)		Non Optimal Only Cost (% of VB)		Bound (% of VB)	
		Both	CPO	CPSat	None	CPO	CPSat	CPO	CPSat	CPO	CPSat
20/0/3/10	10	0.00	20.00	50.00	30.00	n/a	n/a	100.04	100.11	93.90	99.60
20/0/3/20	10	60.00	10.00	0.00	30.00	100.00	533.39	100.00	100.07	99.86	99.00
20/0/3/30	10	100.00	0.00	0.00	0.00	100.00	580.37	n/a	n/a	n/a	n/a
20/0/3/50	10	100.00	0.00	0.00	0.00	111.89	9540.06	n/a	n/a	n/a	n/a
20/0/5/10	10	0.00	0.00	40.00	60.00	n/a	n/a	101.45	100.01	91.41	100.00
20/0/5/20	10	50.00	0.00	0.00	50.00	100.00	555.81	100.55	100.00	97.84	99.82
20/0/5/30	10	90.00	0.00	0.00	10.00	100.00	543.33	100.71	100.00	100.00	97.37
20/0/5/50	10	100.00	0.00	0.00	0.00	103.97	1036.94	n/a	n/a	n/a	n/a
20/0/10/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.92	100.83	92.11	100.00
20/0/10/20	10	10.00	0.00	0.00	90.00	100.00	572.31	101.01	100.01	98.77	99.80
20/0/10/30	10	90.00	0.00	0.00	10.00	100.00	2135.78	100.00	100.30	100.00	99.54
20/0/10/50	10	100.00	0.00	0.00	0.00	100.00	205.64	n/a	n/a	n/a	n/a
30/0/3/10	10	0.00	0.00	40.00	60.00	n/a	n/a	100.05	100.44	91.34	100.00
30/0/3/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.35	100.37	99.66	100.00
30/0/3/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.51	100.13	99.97	99.92
30/0/3/50	10	70.00	10.00	0.00	20.00	100.00	3960.23	100.13	100.00	100.00	99.35
30/0/5/10	10	0.00	0.00	40.00	60.00	n/a	n/a	100.25	100.47	90.19	100.00
30/0/5/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.50	100.72	98.96	99.92
30/0/5/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.28	100.45	99.84	99.75
30/0/5/50	10	90.00	10.00	0.00	0.00	100.00	588.82	100.00	100.55	100.00	100.00
30/0/10/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.45	101.83	95.83	98.46
30/0/10/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.23	101.30	99.53	99.99
30/0/10/30	10	10.00	10.00	0.00	80.00	100.00	3394.24	100.27	100.34	99.74	99.94
30/0/10/50	10	100.00	0.00	0.00	0.00	107.72	14094.19	n/a	n/a	n/a	n/a
50/0/3/10	5	0.00	0.00	0.00	100.00	n/a	n/a	100.00	101.50	98.82	95.84
50/0/3/20	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/3/30	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/3/50	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/5/10	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/5/20	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/5/30	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/5/50	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/10/10	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/10/20	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/10/30	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
50/0/10/50	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/3/10	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/3/20	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/3/30	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/3/50	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/5/10	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/5/20	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/5/30	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/5/50	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/10/10	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/10/20	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/10/30	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/10/50	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a